

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET G — FINAL MASTERY & GRAND CHALLENGE TEST (60 MINS)

CHAPTER 07: WORK, ENERGY & SIMPLE MACHINES

TIME ALLOWED: 60 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-G-07

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET G - FINAL MASTERY & GRAND CHALLENGE TEST):

1. The question paper comprises **25 High-Tier Mastery & Olympiad-Standard Multiple Choice Questions** of **1 mark each**.
2. Tests **In-Depth Conceptual Mastery, Multi-Step Numericals & Rigorous Scientific Deductions**: Recommended target time is **60 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **GRAND CHALLENGE TIER:** This paper features comprehensive multi-step numericals, advanced Olympiad/NTSE-aligned scenarios, and rigorous experimental synthesis designed to achieve absolute 100/100 mastery.

Q.1 A variable force $F = 4x$ (in Newtons) acts on a particle, displacing it from $x = 0$ to $x = 3$ m. What is the work done?

- (A) 12 J (B) 18 J
(C) 36 J (D) 6 J

Q.2 A body of mass 1 kg is whirled in a vertical circle of radius 1 m. What is the minimum speed required at the top of the loop for the string not to go slack ($g = 9.8 \text{ m/s}^2$)?

- (A) 3.13 m/s (B) 4.42 m/s
(C) 9.8 m/s (D) 1.0 m/s

Q.3 What is the minimum speed required at the bottom of the vertical circle for the same body to complete the loop?

- (A) 3.13 m/s (B) 7.0 m/s
(C) 9.8 m/s (D) 5.0 m/s

Q.4 A spring of spring constant $k = 200 \text{ N/m}$ is compressed by 10 cm (0.10 m). What elastic potential energy is stored in the spring?

- (A) 1.0 J (B) 2.0 J
(C) 10 J (D) 20 J

Q.5 A crane lifts an 800 kg car vertically through 15 m in 20 seconds. What is the average power developed by the crane engine ($g = 9.8 \text{ m/s}^2$)?

- (A) 5880 W (B) 6000 W
(C) 117.6 kW (D) 2940 W

Q.6 An electric heater rated 1500 W operates 4 hours daily for 30 days. At a rate of ₹6.00 per unit (kWh), what is the monthly electricity cost?

- (A) ₹540 (B) ₹1080
(C) ₹720 (D) ₹360

Q.7 A bullet of mass 10 g moving at 400 m/s penetrates 20 cm into a wooden target. What is the average resistive force exerted by the wood?

- (A) 4000 N (B) 8000 N
(C) 2000 N (D) 800 N

Q.8 A block and tackle pulley system has 5 pulleys and an efficiency of 80%. What effort is required to lift a load of 1000 N?

- (A) 200 N (B) 250 N
(C) 125 N (D) 500 N

Q.9 If the kinetic energy of a body becomes 9 times its initial value, what happens to its momentum?

- (A) Triples (3 times) (B) 9 times
(C) 81 times (D) $\sqrt{3}$ times

Q.10 Work done by a centripetal force on an object moving along a curved circular path is always:

- (A) Zero (B) Positive
(C) Negative (D) Variable

Q.11 A simple pendulum released from an angle θ has mass m and length L . What is the tension in the string at the lowest point if released from horizontal ($\theta = 90^\circ$)?

- (A) mg (B) $2mg$
(C) $3mg$ (D) $5mg$

Q.12 Two bodies A and B of masses 2 kg and 8 kg have identical linear momentum. Which body has greater kinetic energy?

- (A) Body A (lighter body) (B) Body B (heavier body)
(C) Both have identical kinetic energy (D) Neither has kinetic energy

Q.13 A pump can raise 7200 kg of water per hour to a height of 10 m. What is the power output of the pump ($g = 10 \text{ m/s}^2$)?

- (A) 200 W (B) 720 W
(C) 2000 W (D) 72 kW

Q.14 A lever of length 150 cm has its fulcrum at 30 cm from the load. What is its ideal mechanical advantage?

- (A) 5 (B) 4
(C) 0.25 (D) 2

Q.15 Which of the following is an example of non-conservative force?

- (A) Gravitational force (B) Kinetic friction force
(C) Electrostatic force (D) Ideal spring restoring force

Q.16 What power is required for an automobile of mass 1000 kg to climb a 1 in 20 incline at a constant speed of 15 m/s (neglecting friction, $g = 9.8 \text{ m/s}^2$)?

- (A) 7350 W (B) 14700 W
(C) 735 W (D) 9800 W

Q.17 When a force acts at an acute angle ($0^\circ < \theta < 90^\circ$) to displacement, the work done is:

- | | |
|--------------|--------------|
| (A) Positive | (B) Negative |
| (C) Zero | (D) Maximum |
-

Q.18 A mass of 0.5 kg attached to a spring oscillates on a frictionless table with amplitude 0.2 m and spring constant 50 N/m . What is the maximum speed of the mass?

- | | |
|-----------------------|-----------------------|
| (A) 1.0 m/s | (B) 2.0 m/s |
| (C) 4.0 m/s | (D) 0.5 m/s |
-

Q.19 The commercial energy consumption of a factory is 500 kWh. In Joules, this is:

- | | |
|---------------------------------|---------------------------------|
| (A) $1.8 \times 10^9 \text{ J}$ | (B) $5 \times 10^5 \text{ J}$ |
| (C) $3.6 \times 10^6 \text{ J}$ | (D) $1.8 \times 10^6 \text{ J}$ |
-

Q.20 A body is moved along a closed loop back to its initial position in a gravitational field. Total work done by gravity is:

- | | |
|--------------|--------------|
| (A) Zero | (B) Positive |
| (C) Negative | (D) mgh |
-

Q.21 What is the mechanical advantage of a pair of pliers used to cut wire if effort arm is 12 cm and load arm is 2 cm?

- | | |
|--------|----------|
| (A) 6 | (B) 0.16 |
| (C) 24 | (D) 10 |
-

Q.22 A tennis ball dropped from 2.0 m rebounds to 1.28 m. What percentage of its mechanical energy was dissipated on impact?

- | | |
|---------|---------|
| (A) 36% | (B) 64% |
| (C) 28% | (D) 72% |
-

Q.23 A force $\vec{F} = 3\hat{i} + 4\hat{j} \text{ N}$ displaces an object through $\vec{s} = 2\hat{i} + 5\hat{j} \text{ m}$. What is the work done?

- | | |
|----------|----------|
| (A) 26 J | (B) 15 J |
| (C) 7 J | (D) 30 J |
-

Q.24 What happens to the potential energy of a system when two like positive electric charges are brought closer together?

- | | |
|----------------------|------------------|
| (A) Increases | (B) Decreases |
| (C) Remains constant | (D) Becomes zero |
-

Q.25 If an engine does 1000 J of useful work in 2 seconds, its power is:

- | | |
|-----------|------------|
| (A) 500 W | (B) 2000 W |
| (C) 250 W | (D) 1000 W |
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CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET G)

01	☐ A ☐ B ☐ C ☐ D	02	☐ A ☐ B ☐ C ☐ D	03	☐ A ☐ B ☐ C ☐ D	04	☐ A ☐ B ☐ C ☐ D	05	☐ A ☐ B ☐ C ☐ D
06	☐ A ☐ B ☐ C ☐ D	07	☐ A ☐ B ☐ C ☐ D	08	☐ A ☐ B ☐ C ☐ D	09	☐ A ☐ B ☐ C ☐ D	10	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D	12	☐ A ☐ B ☐ C ☐ D	13	☐ A ☐ B ☐ C ☐ D	14	☐ A ☐ B ☐ C ☐ D	15	☐ A ☐ B ☐ C ☐ D
16	☐ A ☐ B ☐ C ☐ D	17	☐ A ☐ B ☐ C ☐ D	18	☐ A ☐ B ☐ C ☐ D	19	☐ A ☐ B ☐ C ☐ D	20	☐ A ☐ B ☐ C ☐ D
21	☐ A ☐ B ☐ C ☐ D	22	☐ A ☐ B ☐ C ☐ D	23	☐ A ☐ B ☐ C ☐ D	24	☐ A ☐ B ☐ C ☐ D	25	☐ A ☐ B ☐ C ☐ D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & STEP-BY-STEP EXPLANATIONS (SET G)

Q. No.	Correct	Rigorous Scientific Solution & Formula Derivation
Q.1	(B)	$\$W = \int_0^3 4x \, dx = [2x^2]_0^3 = 2(3)^2 = 2 \times 9 = 18 \text{ J}.$
Q.2	(A)	$\$v_{\text{top}} = \sqrt{gr} = \sqrt{9.8 \times 1} \approx 3.13 \text{ m/s}.$
Q.3	(B)	$\$v_{\text{bottom}} = \sqrt{5gr} = \sqrt{5 \times 9.8 \times 1} = \sqrt{49} = 7.0 \text{ m/s}.$
Q.4	(A)	$\$E_p = \frac{1}{2} k x^2 = 0.5 \times 200 \times (0.10)^2 = 100 \times 0.01 = 1.0 \text{ J}.$
Q.5	(A)	$\$P = \frac{mgh}{t} = \frac{800 \times 9.8 \times 15}{20} = \frac{117600}{20} = 5880 \text{ W} (\$5.88 \text{ kW}).$
Q.6	(B)	Energy $\$= 1.5 \text{ kW} \times 4 \text{ h} \times 30 = 180 \text{ kWh}.$ Cost $\$= 180 \times 6 = \text{₹}1080.$
Q.7	(A)	Work-energy theorem: $\$F \times s = \frac{1}{2}mv^2 \Rightarrow F \times 0.20 = 0.5 \times 0.010 \times (400)^2 = 0.005 \times 160000 = 800 \text{ J} \Rightarrow F = 800/0.20 = 4000 \text{ N}.$
Q.8	(B)	$\$\text{VR} = 5\$.$ $\$\text{MA} = \eta \times \text{VR} = 0.80 \times 5 = 4.0\$.$ Effort $\$= \frac{\text{Load}}{\text{MA}} = \frac{1000}{4.0} = 250 \text{ N}.$
Q.9	(A)	$\$p = \sqrt{2m E_k}.$ If $\$E_k$ increases by 9 times, $\$p$ increases by $\sqrt{9} = 3$ times.
Q.10	(A)	Centripetal force is perpendicular to the displacement at every point along the circular trajectory ($\$W = 0\$).$
Q.11	(C)	Velocity at bottom: $\$v = \sqrt{2gL}.$ Tension $\$T = mg + \frac{mv^2}{L} = mg + \frac{m(2gL)}{L} = 3mg.$
Q.12	(A)	$\$E_k = \frac{p^2}{2m}.$ For equal $\$p$, kinetic energy is inversely proportional to mass; the lighter mass has greater $\$E_k.$
Q.13	(A)	$\$P = \frac{mgh}{t} = \frac{7200 \times 10 \times 10}{3600 \text{ s}} = 200 \text{ W}.$
Q.14	(B)	Load arm $\$= 30 \text{ cm}.$ Effort arm $\$= 150 - 30 = 120 \text{ cm}.$ $\$\text{MA} = \frac{\text{Effort arm}}{\text{Load arm}} = \frac{120}{30} = 4\$.$
Q.15	(B)	Friction dissipates mechanical energy into heat; its work depends on path length, making it non-conservative.
Q.16	(A)	$\$F = mg \sin \theta = 1000 \times 9.8 \times (1/20) = 490 \text{ N}.$ Power $\$P = Fv = 490 \times 15 = 7350 \text{ W} (\$7.35 \text{ kW}).$
Q.17	(A)	For $\$0 \leq \theta < 90^\circ,$ $\$\cos \theta > 0,$ making work done positive.
Q.18	(B)	$\$\frac{1}{2}mv_{\text{max}}^2 = \frac{1}{2}kA^2 \Rightarrow v_{\text{max}} = A\sqrt{k/m} = 0.2\sqrt{50/0.5} = 0.2\sqrt{100} = 0.2 \times 10 = 2.0 \text{ m/s}.$
Q.19	(A)	$\$500 \text{ kWh} = 500 \times 3.6 \times 10^6 \text{ J} = 1.8 \times 10^9 \text{ J} (1.8 \text{ GigaJoules}).$
Q.20	(A)	Gravity is a conservative force; the net work done along any closed path is zero.
Q.21	(A)	$\$\text{MA} = \frac{\text{Effort arm}}{\text{Load arm}} = \frac{12 \text{ cm}}{2 \text{ cm}} = 6\$.$
Q.22	(A)	Energy retained $\$= \frac{1.28}{2.00} = 64\%.$ Energy dissipated $\$= 100 - 64 = 36\%.$
Q.23	(A)	$\$W = \vec{F} \cdot \vec{s} = (3 \times 2) + (4 \times 5) = 6 + 20 = 26 \text{ J}.$
Q.24	(A)	Work must be done against electrostatic repulsive forces to bring like charges closer, increasing electrostatic potential energy.
Q.25	(A)	$\$P = W/t = 1000/2 = 500 \text{ W}.$